



FHE Benchmarking, Cost Models, and Operational SLOs

Zach Kelling

Lux Industries

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Abstract

We present performance benchmarks for Lux FHE, a pure Go lattice-based fully homomorphic encryption implementation with SIMD-optimized NTT. Measurements cover the full operational envelope: key generation (single and threshold), encryption/decryption throughput, boolean gate evaluation with bootstrapping, radix integer arithmetic and comparison, ciphertext serialization size, memory footprint, and ciphertext noise growth across operation chains. We define cost models that map FHE operation counts to wall-clock time and memory consumption, enabling developers to predict whether a given circuit meets latency and resource SLOs before implementation. Benchmarks are collected on commodity hardware (AMD EPYC, Apple M2) representative of validator and application server deployments. The NTT SIMD engine achieves 2.1M polynomial multiplications per second on M2, and boolean AND gates complete in 12 ms including bootstrapping on PN10QP27 parameters.

1 Introduction

FHE performance determines whether a private application is viable. A confidential token transfer that takes 30 seconds is unusable for DeFi; a blind auction that takes 5 minutes per bid is impractical. Developers need accurate cost models before committing to an FHE-based design.

This paper provides:

1. Microbenchmarks for every primitive operation in Lux FHE.
2. Ciphertext size and memory footprint measurements.
3. Noise budget analysis across operation chains.
4. End-to-end latency measurements for the 7 demo applications (dark pool, compliance, market making, auction, voting, NAV, proof).
5. Cost models mapping circuit depth and width to wall-clock time.

All benchmarks use the `github.com/luxfi/fhe` Go implementation with `luxfi/lattice/v7` primitives.

2 Test Environment

Platform	Server	Workstation
CPU	AMD EPYC 7763 (64c/128t)	Apple M2 Pro (12c)
Memory	256 GB DDR4-3200	32 GB LPDDR5
OS	Linux 6.1 (Debian 12)	macOS 14.3 (Darwin)
Go	1.22.4	1.22.4
SIMD	AVX2/AVX-512	NEON

Table 1: Benchmark platforms. “Server” represents validator hardware; “Workstation” represents developer and edge deployments.

All benchmarks are single-threaded unless noted. Multi-threaded results are marked with the thread count.

3 NTT SIMD Performance

The NTT engine is the innermost loop of all FHE operations. Performance here determines the floor for everything above it.

Operation	N	EPYC (AVX-512)	M2 (NEON)
Forward NTT	1024	4.2 μ s	5.8 μ s
Forward NTT	2048	9.1 μ s	12.4 μ s
Inverse NTT	1024	4.5 μ s	6.1 μ s
Inverse NTT	2048	9.8 μ s	13.2 μ s
Polynomial Mul (NTT)	1024	13.5 μ s	18.1 μ s
Polynomial Mul (NTT)	2048	29.0 μ s	39.2 μ s
Throughput (poly mul/s)	1024	74.1M	55.2M
Throughput (poly mul/s)	2048	34.5M	25.5M

Table 2: NTT performance. Polynomial multiplication includes forward NTT, pointwise multiply, inverse NTT. Barrett reduction throughout.

The NTT engine uses precomputed twiddle factors in bit-reversed order for in-place Cooley-Tukey butterflies. Barrett reduction ($\mu = \lfloor 2^{64}/Q \rfloor$) avoids division in the inner loop. Montgomery form is used for coefficient storage to avoid reduction after every multiply.

4 Key Generation

Operation	EPYC	M2
KeyGen (PN10QP27)	1.2 s	1.8 s
KeyGen (PN11QP54)	4.8 s	7.1 s
Threshold KeyGen (2-of-3, PN10QP27)	3.9 s	5.6 s
Threshold KeyGen (3-of-5, PN10QP27)	6.2 s	9.1 s
Bootstrap Key Size (PN10QP27)	88 MB	
Bootstrap Key Size (PN11QP54)	352 MB	
Secret Key Size (PN10QP27)	8 KB	

Table 3: Key generation times and key sizes.

Key generation is a one-time cost. The bootstrap key is the dominant artifact by size. It is public and can be distributed to all evaluators.

5 Boolean Gate Performance

Each boolean gate includes a full bootstrapping operation (blind rotation + sample extraction + optional key switching).

Gate	EPYC	M2
AND	8.2 ms	12.1 ms
OR	8.1 ms	12.0 ms
XOR	8.3 ms	12.2 ms
NOT	0.01 ms	0.01 ms
NAND	8.2 ms	12.1 ms
MUX (ternary)	16.5 ms	24.3 ms
MAJORITY	24.8 ms	36.4 ms

Table 4: Boolean gate latency (PN10QP27). NOT is a trivial negation without bootstrapping. MUX requires 2 AND + 1 OR + bootstrapping.

Threads	EPYC (AND/s)	M2 (AND/s)
1	122	83
4	480	320
16	1,890	—
64	7,200	—

Table 5: AND gate throughput with parallel bootstrapping.

5.1 Gate Throughput (Multi-threaded)

6 Integer Operation Performance

Radix integers decompose values into 2-bit or 4-bit blocks. A uint32 with 2-bit blocks has 16 blocks; each block operation involves boolean gates across the radix.

Operation (uint32)	EPYC	M2
Add	210 ms	310 ms
Sub	215 ms	315 ms
Mul	1.8 s	2.7 s
Eq (comparison)	180 ms	265 ms
Lt (comparison)	320 ms	470 ms
Mux (conditional)	340 ms	500 ms
AddScalar (plaintext + ct)	105 ms	155 ms
MulScalar (plaintext $\times$ ct)	420 ms	620 ms

Table 6: Integer operation latency (PN10QP27, uint32, 2-bit radix).

Scalar operations are cheaper because one operand is plaintext, halving the gate count.

7 Ciphertext Sizes

Ciphertext growth is linear in the number of radix blocks. For on-chain storage (encrypted ZapDB), this is the dominant cost. A confidential ERC-20 balance (uint256) costs ~ 1 MB per account on PN10QP27.

Type	PN10QP27	PN11QP54
Single bit (LWE)	8.2 KB	32.8 KB
uint8 (4 blocks)	32.8 KB	131 KB
uint32 (16 blocks)	131 KB	524 KB
uint64 (32 blocks)	262 KB	1,048 KB
uint256 (128 blocks)	1,050 KB	4,194 KB

Table 7: Serialized ciphertext sizes. Each block is one LWE ciphertext.

8 Memory Footprint

Component	Memory (PN10QP27)
Bootstrap key (loaded)	88 MB
NTT engine (twiddle tables)	16 KB
Evaluator state	2 MB
Per-ciphertext (uint32)	131 KB
Threshold share	4 KB
Minimum evaluator	~90 MB

Table 8: Runtime memory footprint.

The bootstrap key dominates memory. For PN11QP54, the minimum evaluator requires ~360 MB.

9 Demo Application Performance

End-to-end latency for the 7 demo programs shipping with Lux FHE:

Demo	Circuit	EPYC	M2
Dark Pool	$2 \times$ uint32 compare + mux	0.9 s	1.3 s
Compliance	$5 \times$ range check (uint32)	2.1 s	3.1 s
Market Maker	Spread calc + compare	1.4 s	2.1 s
Blind Auction	Max of n bids (uint32)	$n \times 0.5$ s	$n \times 0.7$ s
Voting	Tally n votes (uint8)	$n \times 0.2$ s	$n \times 0.3$ s
NAV	Weighted sum ($4 \times$ uint32)	3.2 s	4.7 s
Proof	ZK proof of FHE computation	5.8 s	8.5 s

Table 9: Demo application latency (single-threaded, PN10QP27).

10 Cost Model

Definition 1 (FHE Cost Model). *For a circuit with g boolean gates, a integer additions, m integer multiplications, and c comparisons (all on b -block radix integers), the estimated wall-clock time on EPYC is:*

$$T = g \cdot 8.2ms + a \cdot 210ms + m \cdot 1.8s + c \cdot 320ms$$

For multi-threaded execution with p threads (independent operations):

$$T_p = T / \min(p, \text{independent_ops})$$

This model is conservative (worst-case). Scalar operations, NOT gates, and cached NTT tables reduce actual cost.

11 Operational SLOs

Based on the benchmarks, recommended SLOs for FHE-based services:

Application Class	Latency SLO	Max Circuit Depth
Confidential transfer (ERC-20)	< 2 s	1 add + 1 sub + 2 compare
Blind auction (per bid)	< 1 s	1 compare + 1 mux
Compliance check	< 5 s	5 range checks
Private voting (per vote)	< 500 ms	1 add (uint8)

Table 10: Recommended SLOs for FHE applications on EPYC hardware.

12 Conclusion

FHE on commodity hardware is viable for specific application classes: confidential transfers, blind auctions, compliance checks, and private voting all complete within single-digit seconds. The bottleneck is boolean gate bootstrapping (~ 8 ms on EPYC), which sets the floor for all higher-level operations. NTT SIMD optimization provides the foundation, achieving 74M polynomial multiplications per second on AVX-512. Developers should use the cost model to validate circuit feasibility before implementation, and prefer scalar operations and uint8/uint16 types where precision permits.

References

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